

Akilesh Thuniki

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PROFESSIONAL EXPERIENCE

Cloud Architect and DevOps Engineer with 5+ years of experience designing and optimizing secure, scalable, and highly available cloud infrastructure on AWS. Proven success leading cloud-native transformations, implementing infrastructure as code with Terraform, and streamlining CI/CD using GitHub Actions, GitLab, Jenkins, and ArgoCD. Skilled in Kubernetes, ECS, event-driven microservices, and GitOps, with strong focus on automation, reliability, and cost efficiency. Delivered 99.99% uptime, 35% AWS cost savings, and 70% faster deployments through performance tuning, resilient design, and proactive monitoring with Prometheus, Datadog, Grafana, ELK, Splunk, and CloudWatch. Known for fostering DevSecOps culture, leading cloud migrations, and mentoring teams to align technology with business strategy.

TECHNICAL SKILLS

Cloud & Infrastructure: AWS (EC2, S3, Lambda, API Gateway, Cognito, SNS, SQS, IAM, CloudWatch, Shield, WAF, Backup), Kubernetes (EKS), ECS, Docker, Helm, Terraform, AWS CloudFormation, VPC, Subnets, NACLs, Security Groups, Load Balancers, Auto Scaling

DevOps & CI/CD: GitHub Actions, GitHub Enterprise, GitLab CI, Jenkins, ArgoCD Flux, GitOps, Ansible, Chef, Puppet, Prometheus, Grafana, Datadog, CloudWseratch, Splunk, Elasticsearch

Programming & Automation: Python, Golang, Java, TypeScript, Bash, Automation scripting, workflow orchestration, API integrations

Data Engineering & Analytics: AWS Data Lake (S3, Glue, Athena), AWS EMR, Apache Spark, Redshift, Snowflake, ETL/ELT Pipelines, Data Catalog, QuickSight, Tableau, Data Governance

Databases & Storage: MySQL, PostgreSQL, DynamoDB, MongoDB, Redis

Messaging & Event-Driven Architectures: Kafka, ActiveMQ, SNS, SQS, Kinesis, EventBridge

Agile & Project Management: Scrum, Kanban, Jira, Confluence, ServiceNow, ClickUP

WORK EXPERIENCE

ShadowRock | USA

Cloud Architect | Jan 2025 – Present

- Architected a cloud-native serverless platform on AWS using Lambda, API Gateway, and Step Functions, reducing infrastructure overhead by 50% while improving scalability, resilience, and fault tolerance for business-critical APIs.
- Established GitOps practices with Flux and GitLab CI/CD for infrastructure automation using Terraform, reducing deployment time by 60% and enabling zero-downtime deployments across environments.
- Designed and implemented a robust zero-trust security model with AWS Cognito, WAF, and Shield, achieving 100% compliance with industry security standards and enabling MFA enforcement across all customer-facing applications supporting digital transformation initiatives.
- Engineered a resilient event-driven API ecosystem using SNS, SQS, and Kinesis for asynchronous processing, improving system reliability by 40% and reducing API latency by 65%.
- Optimized AWS resource usage via right-sizing, auto-scaling, and spot instances, achieving 35% cost savings while implementing FinOps and establishing SLOs and error budgets across multiple development teams.
- Led adoption of AWS Well-Architected principles across multiple workloads, ensuring 99.99% uptime through multi-AZ architecture, automated recovery, and comprehensive CloudWatch monitoring with SLIs/SLOs in an ITIL-aligned ServiceNow change/incident management environment.

BulkMagic| USA

DevOps Engineer | May 2024 – Jan 2025

- Automated CI/CD pipelines with GitLab CI/CD and Kubernetes (ECS, EKS) for microservices architecture, improving deployment efficiency by 35% and reducing release failures by 45% with automated testing and incident protocols.
- Developed Java and Python-based automation scripts to streamline repetitive workflows and maintenance tasks, reducing manual effort by 30% across 5 development teams and improving developer productivity and code quality.
- Integrated Splunk and Prometheus with custom alerting and dashboards for monitoring business-critical APIs, detecting issues 30% faster, increasing uptime to 99.95% through SRE practices, and providing insights for performance optimization.
- Optimized REST APIs (Python, Flask, Java) through caching, pagination, validation, and pooling, improving backend performance by 35%, reducing response time to under 100ms, and enhancing user experience across distributed systems.
- Built scalable event-driven architectures using Kafka and SQS with dead-letter queues, boosting system responsiveness by 40% and enabling asynchronous processing of 10M+ events across distributed systems with high availability and resilience.
- Automated infrastructure provisioning (Terraform, Ansible) with modular templates, reducing environment setup time by 50%, ensuring configuration consistency, and simplifying team onboarding while supporting 24/7 operations in compliance with ITIL-based incident and change processes.

Capgemini Technology Services | Hyderabad, IN

DevOps Engineer | Jun 2021 – Jul 2023

- Led DevOps digital transformation for multiple Fortune 500 clients, improving deployment reliability by 85% and reducing time-to-market by 40% through SRE practices and establishing continuous improvement processes with excellence.
- Developed custom Golang, Java, and TypeScript automation scripts with robust error handling, reducing infrastructure provisioning time from days to minutes, eliminating 90% of manual errors, and supporting API ecosystem scalability.
- Successfully migrated 200+ legacy applications to containerized microservices using EKS with zero data loss, enhancing scalability by 60% and reducing downtime by 75% while maintaining high availability and backward compatibility.
- Implemented high-throughput, fault-tolerant Kafka message brokers for real-time event processing for handling 5M+ daily transactions with sub-second latency, 99.99% reliability, and resilience through performance optimization.
- Built and optimized GitHub Actions CI/CD pipelines for test automation, accelerating deployments by 300%, enabling daily releases across product teams, and reducing failures by 70% through comprehensive quality and incident management.
- Enhanced system observability with Elasticsearch, Splunk, and centralized logging, reducing MTTR by 65% and improving incident response time from hours to minutes through custom dashboards, proactive alerting, and SLI tracking
- Deployed blue-green release strategies for production workloads, minimizing customer disruptions during 300+ successful deployments and decreasing rollback time from hours to minutes with flag management and error budget monitoring.

Tata Consultancy Services (TCS) | Bangalore, IN

Cloud Engineer | Jul 2019 – Apr 2021

- Architected AWS networking infrastructure (VPC, subnets, security groups, ELB) for 30+ critical business applications across multiple regions achieving 99.9% uptime and 70% fewer security incidents throughout the enterprise.
- Automated cloud provisioning with Terraform and CloudFormation templates and modules, cutting deployment time from days to minutes with 100% configuration consistency and significantly improved developer productivity across all teams.
- Built and maintained highly scalable Jenkins CI/CD pipelines for 50+ microservices and Lambda functions with automated testing enabling 20+ daily deployments and 80% faster releases through sophisticated automation frameworks.
- Deployed Docker and Kubernetes (ECS, EKS) solutions across AWS regions with auto-scaling policies that improved resource utilization by 40% and handled 3x traffic spikes during peak hours without performance degradation.
- Implemented CloudWatch, CloudTrail, and Grafana monitoring with custom dashboards and alerting mechanisms, reducing resolution time by 65% and enhancing security visibility for operational teams throughout the organization.
- Optimized IAM roles and access policies using least-privilege principles and security audits with automated compliance scanning, reducing vulnerabilities by 85% and ensuring SOC2/HIPAA compliance for enterprise customers.

EDUCATION

University of Utah

Master of Science in Information Systems

Utah, USA

Aug 2024

Jawaharlal Nehru Technological University

Bachelor of Technology in Mechanical Engineering

Hyderabad, IN

Jun 2019

PROJECTS

Infrastructure Migration on AWS & Service Enhancement

- Led a large-scale cloud migration with SRE principles, ensuring 99.99% uptime, high availability, and reducing costs by 30% through performance optimization and operational excellence.
- Strengthened security using AWS IAM, Shield, WAF, and Patch Manager, mitigating 70% of potential threats and supporting business-critical APIs with resilience.
- Optimized load balancing with SLO monitoring to ensure stability during peak traffic while maintaining error budgets.

Enterprise CI/CD Pipeline Modernization

- Built a GitOps-driven CI/CD pipeline supporting API ecosystem automation, automating deployments for 50+ microservices with distributed systems architecture.
- Reduced deployment failures by 75% with automated testing, rollback strategies, and incident management protocols.
- Integrated GitLab CI/CD & Flux, improving deployment speed, reliability, and operational excellence across REST APIs.

Cloud Cost Optimization Initiative

- Led an AWS cost reduction initiative, cutting cloud spend by 35% through resource scheduling and right-sizing.
- Developed real-time cost monitoring dashboards (Grafana, CloudWatch) for improved visibility.

CERTIFICATIONS

- AWS Certified Developer
- Agile Software Development by the University of Minnesota
- QA Automation Engineer – Level 1 by Capgemini
- AWS Cloud Practitioner Essentials by Amazon Web Services